

Institute of Actuaries of India

Subject SP7 – General Insurance Reserving and Capital Modeling

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Fidelity guarantee insurance protects a company against dishonest actions of its employees such as fraud/theft /embezzlement which cause financial loss or reputational damage to the company. Directors' and Officers' liability insurance provides coverage to directors and officers of the company for claims made by third parties against directors and/or officers of the company for losses caused by wrongful acts performed in their capacity as directors and/or officers.

The main difference is the entity that will make a claim under the policy or whether coverage is for illegal versus inappropriate acts.

[1/2 mark each for definition of Fidelity guarantee insurance and Directors'/Officers' Liability insurance and 1 mark for stating the differences]

[2]

- ii) For a claim to be payable the policyholder must show that the insured is liable.
The claim amount distribution is typically positively skewed but depends on the class of business e.g. for Motor there may be small claims for property damage and large claims for bodily injury. [1]
Claim amounts depend on the generosity or otherwise of compensation awards by the legislations governing a jurisdiction [0.5]
Liability class of insurance is exposed to various types of delay e.g. reporting and settlement delay, risk of reopening claims and latent claims [0.5]
The claim amount may be affected by consideration of contributory negligence.
Frequency depends on the class of business [0.5]
There can be accumulation of claims e.g. a defective product triggering an influx of product liability claims [0.5]
Most claims may be contested in court. This increases settlement delays and exposes claims to judicial inflation [0.5]
Most lines of liability insurance are very susceptible to super imposed inflation [0.5]

[Marks to be awarded for alternative points]

[4]

- iii) Occurrence policies provide cover for losses incurred by the insured during the period of the policy. Claims made policies provide cover for losses reported by the insured during the policy period irrespective of when the policy was written and claim occurred. [1.5]
This means that when the market moves to claims made basis of underwriting policies, many losses reported will relate to policies underwritten on the old basis with potentially different policy conditions, exclusions and underwriting risk acceptance criteria. Claims that occurred under the old system will need to be excluded from the new, particularly in the case of classes and jurisdictions prone to latent claims.
This means that claims made policies underwritten in the early days of transition, will have significant exclusions and will be subject to substantial discounts on premium that will need to be withdrawn over time. [2]
This means that the total premium income of the sector will be depressed, with implications for expense loadings and commission rates. [0.5]

If the whole market does not transition to the claims made basis of underwriting, then policy exclusions will need to be tailored to each individual insured's circumstances, with implications for underwriting procedures. [0.5]

Claims made policies will not provide cover for losses reported after the policy expiry date. This will create a need for run-off policies. For example, professional people who retire may still be sued for their actions while still in practice and will need this cover; policies will need to be developed to meet this need. [1]

There will also be problems with businesses that have closed, become bankrupt or been the subject of a merger/acquisition. Legal restrictions may therefore be placed on the cover that may be offered, restricting the insurers' ability to fully transition to the new basis of underwriting. [1]

There will be issues from a pricing and reserving perspective as the development period is much shorter for claims made policies than for losses occurring. [1]

There may be issues with outwards reinsurance e.g. ensuring no gaps [0.5]

Policyholder communication required e.g. to ensure policyholders are aware of the change to their coverage. [0.5]

[Marks to be awarded for alternative solutions]

[8]

- iv) Super imposed inflation can be described as claims inflation above and beyond what would be expected from developing claims experience and relevant indices of economic inflation. [2]

Key drivers of superimposed inflation are:

- Increasing litigation
- Broader definitions of liability
- More plaintiff-friendly legal decisions
- Larger compensation jury awards
- General anti-corporate sentiment
- Social media and advertising and other actions by the plaintiff's bar
- Litigation funding
- Politics and regulation and government actions

[0.5 marks for any relevant point subject to a maximum of 4]

[6]

[20 Marks]

Solution 2:

- i) Assumptions:

- AY2017 is fully run-off
- Run-off pattern is same for each AY
- Future experience is expected to be consistent with past experience
- Inflation assumption (future is similar to past or any other sensible assumption consistent with the calculations)
- The triangulation relates to net paid claims

AY	Development period				2	3	Ultimate	Latest	Reserve
	1	2	3	4					
2017	50	200	250	300			300	300	0
2018	125	200	250				300	250	50
2019	75	100				125	150	100	50
2020	100				200	250	300	100	200
Dev factor	2	1.25	1.2	1			1050		300
Cumulative Dev Factor	3	1.5	1.2	1					

[1 mark for assumptions, 1 mark for development factor calculations, 1 mark for reserves]
 [Marks available for other correct sensible calculations eg for AY 2019, if ultimate is calculated using BF or ELR methods with reasonable explanation]
[3]

ii) Discount rate calculation = 6% + 4% = 10%

[1 mark, marks available for using 6% with reasonable explanation]

AY	Accounting Year			Total
	2021	2022	2023	
2017				0
2018	50			50
2019	25	25		50
2020	100	50	50	200
Total	175	75	50	300
Discount factor	1.1	1.21	1.331	
Discounted reserves	159	62	38	259

[1 mark for discount factors, 1 mark for discounted reserves]

Expected impact on the balance sheet and P&L:

- The liability for claims reserves will reduce by the amount of discounting (Rs. 41 crores)
- There will be an increase in profits / reduction in losses by 41 crores in the year ending 2020....
-if not done retrospectively, i.e. if 2019 accounts are not restated for discounting impact

- Future investment income will be partially offset by the impact of discounting unwind.

[1 mark each subject to a maximum of 2]

[6]

iii) Assumptions:

- No other expense/income that could impact the ULR
- Actual is in line with expected

	NEP	ULR
2017	500	60%
2018	500	60%
2019	500	30%
2020	500	60%

[0.5 marks for each correct ULR, based on reserves calculated in part (i) and any other assumptions. 0 marks awarded if using discounted reserves for calculating ULR]

[2]

- iv) One of the key assumptions for BCL method is that future experience is similar to the past. However for AY2019 the BCL method seems to underestimate the ultimate. While it is possible that AY2019 had particularly favorable claim experience, this is unlikely to be the case since the insurer is an established insurer, so expected to have stable business.

All the other AYs in the data provided have an ultimate claims cost of Rs 300 crores and a ULR of 60%.

- This could be because of lower claims paid till 31/12/2020 for AY2019.
- There could be delays driven by legal disputes / other claim handling delays.
- We could look at the incurred claims triangle to check if it gives us more reliable estimates.

We could also speak to the underwriters and claims teams to understand their views and ascertain if AY2019 was indeed an exceptional year or if the BCL method is not appropriate here. We could also consider using BF method or ELR method.

Under BF method the Ultimate for AY2019 will be at 200 (or any other suitable calculated estimate). Under ELR method, the Ult for AY2019 will be 300, given the consistency of other AYs and extremely low volatility. ELR seems to be an appropriate method. However, when looking at the claims paid in development period 3 and 4, they are pretty stable at 50 crores each, so BF could provide a better estimate in this case

In general practice, for the most recent quarter/years, due to relatively lesser experience, BF is the usual choice of method.

[0.5 marks for each point subject to a maximum of 7]

- v) GLM bootstrapping steps:
- Define and fit a GLM, obtaining parameters and fitted values for the observed data. [0.5]
 - Calculate the residuals of your fitted model. [0.5]
 - Take a sample from the residuals (this is the 'bootstrapping bit'), and invert these to obtain a set of pseudo-data. [1]
 - Refit the GLM using this pseudo-dataset, to obtain another set of parameters for the model, and another forecast output. [0.5]
 - Repeat steps 3 and 4 many times to derive a forecast output for each pseudo-dataset. This gives a distribution of parameters and outputs. [0.5]
- [3]
- vi) Assumptions:
- No other items in P&L and Balance Sheet apart from information already provided
 - Reserves are booked on non-discounted basis
 - Claims reserves as at 31/12/2020 – same as (i)

Calculation: Assets = Claim reserves + Premium Reserves + Shareholder's funds

Premium reserves = $700 - (200 + 300) = \text{Rs. } 200 \text{ crores}$

[0.5 mark for each assumption, max 1, 1 mark for formula/workings, 1 mark for correct calculation]
[3]

vii)

a) Balance sheet as at 31/12/2020

Assets: 700

Liabilities:

Claim reserves = 300

Premium reserves = 200

Shareholder's funds = 200

[1 mark for BS calculations]

b) Balance sheet as at 31/12/2021:

Assume premium is written evenly during the year

Assume no other expenses / income.

[1 mark for appropriate assumptions]

Assets = $700 + 600 \text{ (NWP)} - 175 \text{ (Claims paid from 2020 reserves)} - 100 \text{ (Claims paid for 2021 NEP)} = 1025$

[1 mark for asset calculation, 1 mark for clear workings, max 2]

Liabilities:

Claim reserves = $25 \text{ (AY2019)} + 100 \text{ (AY2020)} + 100 \text{ (AY2021)}$

AY2021 Ult = $500 \times 60\%$, AY 2021 paid in 2021 = $300/3 = 100$

Claim reserves = 225

[1 mark for each correct AY calculation, 1 mark for total, max 4]

Premium reserves = $600 \text{ (NWP)} / 2 = 300$

[0.5 mark]

Shareholder's funds:
 Opening balance = 200
 + Profits (calculated below) = 300

[0.5 mark]

[8 marks]**viii) UW Profit:**

Income:

NEP = NWP + Change in UPR

= 600 + (200-300) = 500

Expense:

NIC = Paid claims + change in reserves

= 175 + 100 + (225-300) = 200

UW Profit: 500 – 200 = 300

[1.5 marks for NEP and NIC, 1 mark for profit, total 4]

[37 Marks]**Solution 3:**

The protection of policyholder's interest encompasses the timely payout of valid claims in full as per agreed insurance contracts, timely quality of pre & post issuance of policy servicing, renewal guarantee at a fair and equitable price depending upon the product and regulations.

One of the two companies forming part of the merger may adopt a more restrictive approach to claims settlement than the other and it may be its standards that prevail in future, to the disadvantage of the other company's policyholders.

The merger should not create monopolies and restrict fair competition in the market in future.

Anti-selection could take place post-merger if the underwriting acceptance criteria are not consistent across both companies.

Rating criteria and pricing mechanism should not be changed significantly post-merger which would put any company's policyholder at a significant disadvantage.

Policyholders of both companies should be afforded similar or enhanced product coverage post merger to ensure that no policyholder is worse off as a result of the merger.

Integration of systems and processes should not cause disruptions and delay in the execution of services.

The score given by rating agencies pre and post merger should be considered.

The experience and reputation of the management team of the two companies individually and of the merged entity would be of keen interest to the regulator.

Investigations should cover all potential risks to the companies' solvency: insurance risk, credit risk, market risk and operational risk etc.

An independent review of the reserves/provisions of each company to ensure that comparison between the two companies is made on a consistent basis. There should be consideration given to the risk margin incorporated in the reserves as part of the capital assessments but as reserves

constitute a large proportion of any insurer's balance sheet this is likely to be a matter that the actuary will wish to independently verify.

Concentration and accumulation risk should be considered if the two companies underwrite very similar business.

Post-merger, the appropriateness and adequacy of reinsurance arrangements and its impact on policyholder's interests should be considered and the actuary should undertake investigations around it. e.g. run-off arrangements, adverse development covers.

Post-merger tie-ups with garages, hospitals and TPAs should also be considered to make sure policy and claim servicing is unhindered.

The capital position of each company and the merged entity should be assessed independently to gauge adequacy of capital to meet future policyholder obligations.

Professional guidance standards and relevant regulations should be considered at every stage of review

[1 mark for any valid point subject to a maximum of 15]

Solution 4:

- i) The figures may be pre diversification credit or after considering diversification impact. Property catastrophe has a high capital allocation relative to premium written due to higher concentration risk by location. This seems reasonable as exposure to natural disasters (e.g. TC, EQ, Flood and secondary peril losses) means significant cat risk and increased underwriting risk due to the very volatile nature of business. Being short tailed in nature, reserving risk is not significant. Generally, a catastrophe may increase demand for raw materials/labor and can lead to demand surge with potential liquidity risk.
- Medical malpractice is a type of professional indemnity cover and also has quite a high capital requirement as it is likely to be dominated by reserving risk due to the long tailed nature of this business. It is also exposed to market risk stemming from fluctuations in investment income. Further, it is exposed to changes in the legislative environment which may set the precedence for an influx of unforeseen claims.
- General liability is also a long tailed line of business but unlikely to have high reserve requirements due to contract tenure. However, it is a new class of business for the company so it may be allocated more capital for new business risks i.e. uncertainty around pricing, risk from adverse selection and lack of internal historic data etc. These contribute to increased underwriting risk.
- Motor Insurance has been written for a significant number of years. While Motor OD is short tailed, Motor Third Party is a long tailed line of business, so likely to have significant reserve risk due to possible legislative changes and judicial inflation. It may have aggregation with general liability as both exposed to bodily injury claims and super imposed inflation. Motor lines of business have generally lower credit risk due to Section 64 VB.
- Livestock insurance has been written for a very significant number of years and normally has low reserves and fairly stable loss ratios. These reasons contribute to its overall low capital requirement. Some of the amount may be due to political risk
- [2 marks for each line of business, max 10]
- ii) It is normally the case that not all listed products will simultaneously perform badly and it is generally allowed for through diversification credit in a capital model. Often it is done on a top-down basis when allocating capital.

It is unlikely that there is perfect negative correlation and that it will exactly offset. There are also common causes leading to all classes performing badly simultaneously e.g. soft underwriting cycle, rating pressure and a general hard reinsurance market in several lines simultaneously will have an impact.

Legal changes could impact several classes of business simultaneously e.g. the way claims are settled for bodily injury claims.

There could be systematic underpricing and therefore losses in all classes. This may be arising from inadequate data or failure of underwriting controls.

Any inefficiencies in the claims department could impact several classes e.g. inaccurate case reserving or claims leakage from delays in settlement. Therefore company-wide risks (e.g. operational, reputational) need to be distributed across individual lines of business. Actual vs budget premium could drive very different or incorrect capital allocation.

[1 mark for any valid point, max 5]

[15 Marks]

Solution 5:

i) Some reasons that insurer A would want to purchase reinsurance:

To limit its exposure to risk (or spread risk) in respect of

- single risks
- aggregations of single risks
- accumulations

To obtain additional business through reciprocity

To avoid large losses in respect of single large claims, catastrophes

To increase profitability by

- reducing the volatility of results (and so the ability to plan)
- taking advantage of cheap reinsurance
- to enable it to declare profits from outstanding liabilities more quickly

To improve the solvency margin and hence reduce the risk of insolvency

To increase the capacity to accept risk, either:

- singly, ie to enable it to write larger risks, or
- cumulatively, ie to enable it to write more business

To obtain financial assistance to help with

- new business strain
- bolstering free assets
- a merger / acquisition
- any other short-term cashflow needs

To get technical assistance on new or unusual risks or risks in new territories

Regulatory requirement / as a supervisory condition

[0.5 mark for any valid point, max 10]

ii) Factors to be considered when purchasing reinsurance:

- The classes of business written, the size and range of risks and their volatility
- The amount of capital held
- The extent to which accumulation of risk is possible
- The size (total premiums) and diversification of the insurer's portfolio
- The general cost (value for money) and availability of reinsurance

- Alternatives including coinsurance and protection afforded by parent company
- Financial strength of reinsurers
- Regulatory requirements and fiscal system
- The impact of capital management
- The need for technical assistance particularly for classes of business where the insurer has insufficient internal experience/data.
- Risk appetite and tolerance
- Underwriters' influence
- Financial objectives
- The reinsurers' requirement for a minimum retention
- Relationship with reinsurers and brokers.

[0.5 mark for any valid point, max 3]

[13 Marks]
